

Safety: group lead: Zoe ✓ safety lead: Anna G ✓ data lead: Clara
emergency comm: Radio channel 6, TA: Julia 425-289-6138, Matt: 317-674-3437 ✓ great!
site hazards: parking lot, hot springs, bugs, sun, people with dogs ✓
equipment safety: organize cables & try not to trip on them, no people in plane of hammer swing ✓

Objectives: Figure out how the groundwater level changes as you move away from the spring up the hill ✓ this was a great objective for this method

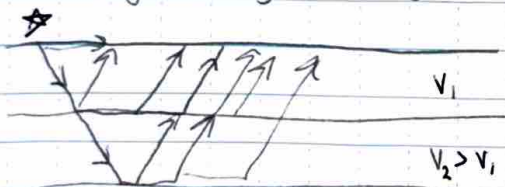
limiting factors: There may be fast-velocity travertine rock overlying slower velocity rock which is not good for refraction method ✓

Methods: awesome !!

near surface seismic is sensitive to changes in the seismic impedance of the ground, which is controlled mainly by the layer velocity, as well as density. the water table would have a high velocity (P-wave), so it would stand out from the unsaturated overlying soil/rock. ✓

theory overview: ✓

for reflection surveys, you can get the velocity and thickness of layers assuming velocity increases as you get deeper



the first few nodes will have the direct wave from near the surface as the first arrival, but later on the head wave that travelled along

the faster layer interface would arrive sooner, and you can get the velocities of each layer by looking at the travel-time curve

nice theory!

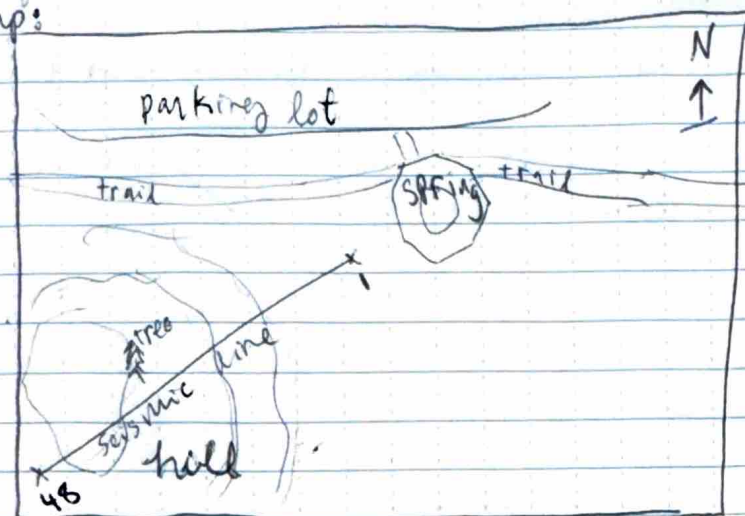
tool to read SEG2 files: geo-giga express ??

Site info: Lithia hot spring (arrival at ~10:30 am)

coords for the spring ✓
40°29'18"N, 106°50'55"W

Conditions: Sunny, light breeze, mostly grassy field, the spring could be a hazard if someone fell in, sun as well could be a hazard. There may be bugs in the grass ✓

map:



yes!!
great map

Survey Parameters:

- 48 geophone array with 1m spacing ✓
- node 1 is closest from the spring ✓
- recording time for each shot: 0.5 sec ✓
- sample freq: every 1/4 ms ✓
- # shots per shot location: 4 at most locs, more in some ✓

Weather stayed sunny with some clouds, minimal wind all day ✓

Potential errors: Vehicle traffic nearby, people/dogs walking, some kind of power line area over the hill, potentially inconsistent hammer shots ✓

another big error may have been poor coupling for the geophones in travertine

Data storage: In one drive, seismic in SEG2 file, GPS in CSV also in folder ✓

hammer shot # loc (m) along line file name header # stacks

1	1	2001	9
2	7.1	2002	6
3	13	2003	4
4	19	2004	4
5	25	2005	4
6	31	2006	4
7	37	2007	4
8	43	2008	4
9	49	2009	7

data lead: Clara ✓

group 5 - ns seismic

for header file info,
make a separate
bullet pt. pls !!

Near Surface Seismic cont.

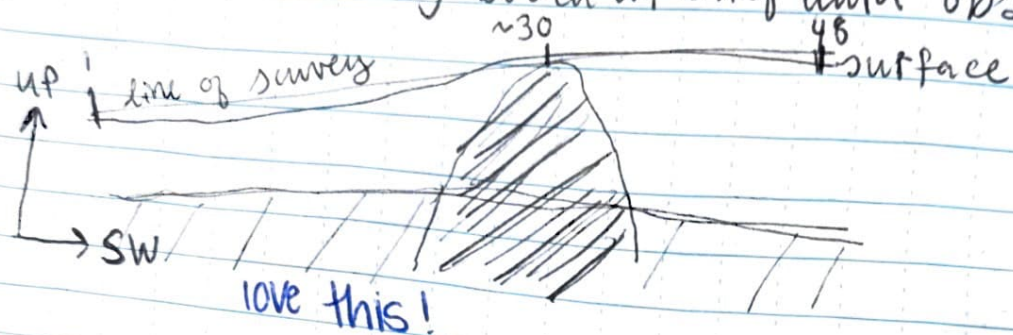
May 19, 2026

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Wrap up:

- Safety: potential concerns: Sun exposure, open hot springs ✓
no accidents/near misses
- We accomplished our task to investigate how the seismic velocities changed as we went out from the spring, we ~~did~~ couldn't find out exactly what was water vs. shale/other things but we found a cool anomaly.
- Challenges: some rocky parts were hard to get nodes in, dead tablet for GPS ✓
- future work: we want to go back with resistivity or GPR to get more constraints on what velocities correspond to what materials. Another seismic line perpendicular & centered on the anomaly could be good to constrain its shape as well nice!!

Hypothesis for velocity based on brief data observations:



- ☐ slowest vel.
- ☒ medium vel.
- ☒ highest vel.

Self Assessment: I learned how to do a hammer seismic. This was great because I had to miss that I was seeing how we could start combining...